Post-Arrest Temperature Management

AC-13 | Adult Cardiac

Quick flow

01 Sustained ROSC [EMT / AEMT / Paramedic]

Open AC-10 • stabilize airway, ventilation, perfusion, rhythm, ECG, and glucose

02 Follows Verbal Commands? [EMT / AEMT / Paramedic]

Yes: continue AC-10 and prevent fever • No: enter temperature-management pathway

03 Measure Temperature [EMT / AEMT / Paramedic]

Use a reliable core or approved temperature method when available • document site and trend

04 Prevent Fever / Exposure [EMT / AEMT / Paramedic]

Remove excessive heat sources • maintain normal environment • prevent further heat loss if cold

05 No Routine Field Cooling [EMT / AEMT / Paramedic]

Do not rapidly infuse cold IV fluids or initiate uncontrolled active cooling

06 Spontaneously Hypothermic? [EMT / AEMT / Paramedic]

Prevent additional heat loss • avoid rapid active rewarming

07 Monitor + Transport [EMT / AEMT / Paramedic]

ECG • SpO_2 • BP/MAP • temperature • $EtCO_2$ with advanced airway • early receiving notification

Overview

- Prehospital temperature-management pathway for an adult with sustained ROSC who remains unresponsive to verbal commands.
- Field priorities are accurate temperature assessment, prevention of fever and further heat loss, avoidance of rapid cold-fluid cooling or rapid rewarming, and uninterrupted post-resuscitation stabilization and transport.

Provider-Level Actions

Temperature management supplements AC-10 and must not delay post-ROSC stabilization or transport.

EMT

- Assess response to verbal commands, obtain and document temperature when equipment is available, and prevent fever, excessive heat exposure, or further heat loss.
- Continue cardiac/AED monitoring, SpO_2, serial blood pressure, glucose assessment, airway support, and frequent reassessment.
- Do not initiate rapid cold-fluid cooling or uncontrolled active surface cooling.

AEMT

- Perform all EMT actions and establish IV access without delaying transport.
- Administer fluids only for the hemodynamic indications and doses in AC-10; do not use chilled crystalloid for temperature control.
- Trend temperature, glucose, blood pressure, rhythm, oxygenation, and response to treatment.

Paramedic

- Perform all prior actions and direct advanced airway, waveform capnography, ventilation, hemodynamic support, serial ECG, and re-arrest preparedness under AC-10.
- Document neurologic responsiveness before sedative or paralytic medication when feasible and report the temperature trend and neurologic findings to the receiving hospital.

- Communicate that an adult remaining unresponsive requires a deliberate receiving-hospital temperature-control strategy.

Indications

- Adult with sustained ROSC after cardiac arrest who remains unresponsive to verbal commands, regardless of arrest location or initial rhythm.

Contraindications

- No separate active prehospital temperature-control pathway is indicated for an adult who follows verbal commands after ROSC; continue AC-10 and prevent fever.
- Routine prehospital cooling with rapid cold-IV-fluid infusion is not indicated.

Assessment | Eligibility and neurologic assessment

- Confirm sustained ROSC and immediately begin or continue AC-10 post-resuscitation care.
- Assess whether the patient follows verbal commands and document the neurologic examination before sedatives or paralytics when feasible.
- Do not make an early neurologic prognosis in the field.

Assessment | Temperature assessment

- Measure temperature as soon as practical using a reliable core or approved device when available.
- Document the measurement site, value, time, environmental exposure, wet clothing, and temperature trend.
- Assess for fever, spontaneous hypothermia, shivering, environmental heat or cold exposure, and treatments performed before EMS arrival.

Assessment | Concurrent post-ROSC monitoring

- Continue cardiac monitoring, defibrillation-pad placement, SpO_2, serial blood pressure/MAP, glucose, and temperature trending.
- Use continuous waveform capnography when an advanced airway is present and reassess airway position after every movement.
- Obtain and transmit a 12-lead ECG, treat hypotension under AC-10, and remain prepared for immediate re-arrest.

Treatment steps

- Open AC-10 and prioritize airway, physiologic ventilation, oxygenation, MAP at least 65 mmHg, glucose assessment, 12-lead ECG, and treatment of the arrest cause.
- If the adult remains unresponsive to verbal commands, measure and trend temperature while preventing fever and avoidable environmental heat exposure.
- Do not routinely initiate active prehospital cooling and do not rapidly infuse cold IV fluids for temperature control.
- If the patient is spontaneously hypothermic, remove wet clothing, insulate from further heat loss, handle gently, and avoid rapid active rewarming.
- Continue continuous monitoring and frequent reassessment; treat re-arrest, dysrhythmia, hypotension, hypoglycemia, airway problems, or seizures under the applicable linked protocol.
- Notify the receiving hospital early and report arrest details, ROSC time, neurologic responsiveness, temperature value/site/trend, airway, $EtCO_2$, ECG, blood pressure, glucose, and treatments.
- Document that the receiving-hospital AHA temperature-control strategy for adults remaining unresponsive is 32°C-37.5°C for at least 36 hours; this is destination guidance and not a field target EMS must achieve.

Warnings / precautions

- Do not let temperature measurement or intervention delay airway management, ventilation, perfusion support, ECG acquisition, destination notification, or transport.
- Do not rapidly infuse cold IV fluids for routine prehospital temperature control.
- Do not use uncontrolled ice packs, immersion, or another active-cooling method that cannot be continuously monitored and regulated.
- Avoid rapid active rewarming of a spontaneously hypothermic post-arrest patient.
- Do not make early neurologic prognostication in the field.

Clinical pearls

- AHA recommends a deliberate hospital temperature-control strategy for every adult who remains unresponsive to verbal commands after ROSC.
- The recommended hospital temperature range is 32°C-37.5°C and the strategy should continue for at least 36 hours while the adult remains unresponsive.
- Prehospital priorities are stabilization, fever prevention, accurate communication, and avoidance of harmful rapid cooling or rewarming.
- A normal initial temperature does not eliminate the need for ongoing fever prevention and hospital temperature control.

Special populations | Spontaneous hypothermia after ROSC

- Prevent further heat loss and avoid rapid active rewarming.
- Open TE-05 when hypothermia contributed to the arrest or remains a primary clinical problem.

Special populations | Responsive after ROSC

- Continue AC-10 care and avoid fever, but do not place a patient who follows verbal commands into the unresponsive post-arrest temperature-control pathway.

Special populations | Post-intubation care

- Open AR-08 for advanced-airway confirmation, ventilation, sedation, analgesia, and reassessment after movement.

Related pathways / contacts

AC-10 Post-Resuscitation Care - Primary pathway for all immediate post-ROSC stabilization.

AC-09 VF / Pulseless VT - Open immediately for re-arrest in VF or pulseless VT.

AC-04 ACS / STEMI - Open for persistent STEMI, acute coronary occlusion, or ongoing ischemia after ROSC.

AR-08 Post-Intubation / BIAD Management - Open for advanced-airway confirmation, ventilation, sedation, and analgesia.

UP-04 Altered Mental Status - Open for glucose below 70 mg/dL or another persistent altered-mental-status cause.

TE-05 Hypothermia - Open when hypothermia contributed to the arrest or remains clinically significant.

Source and references

American Heart Association. 2025 Guidelines for CPR and ECC: Post-Cardiac Arrest Care - Temperature Control After Cardiac Arrest in Adults. | American Heart Association. 2025 Adult Post-Cardiac Arrest Care Algorithm. | Claiborne EMS post-resuscitation, airway, glucose, hypothermia, and rhythm protocols.

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